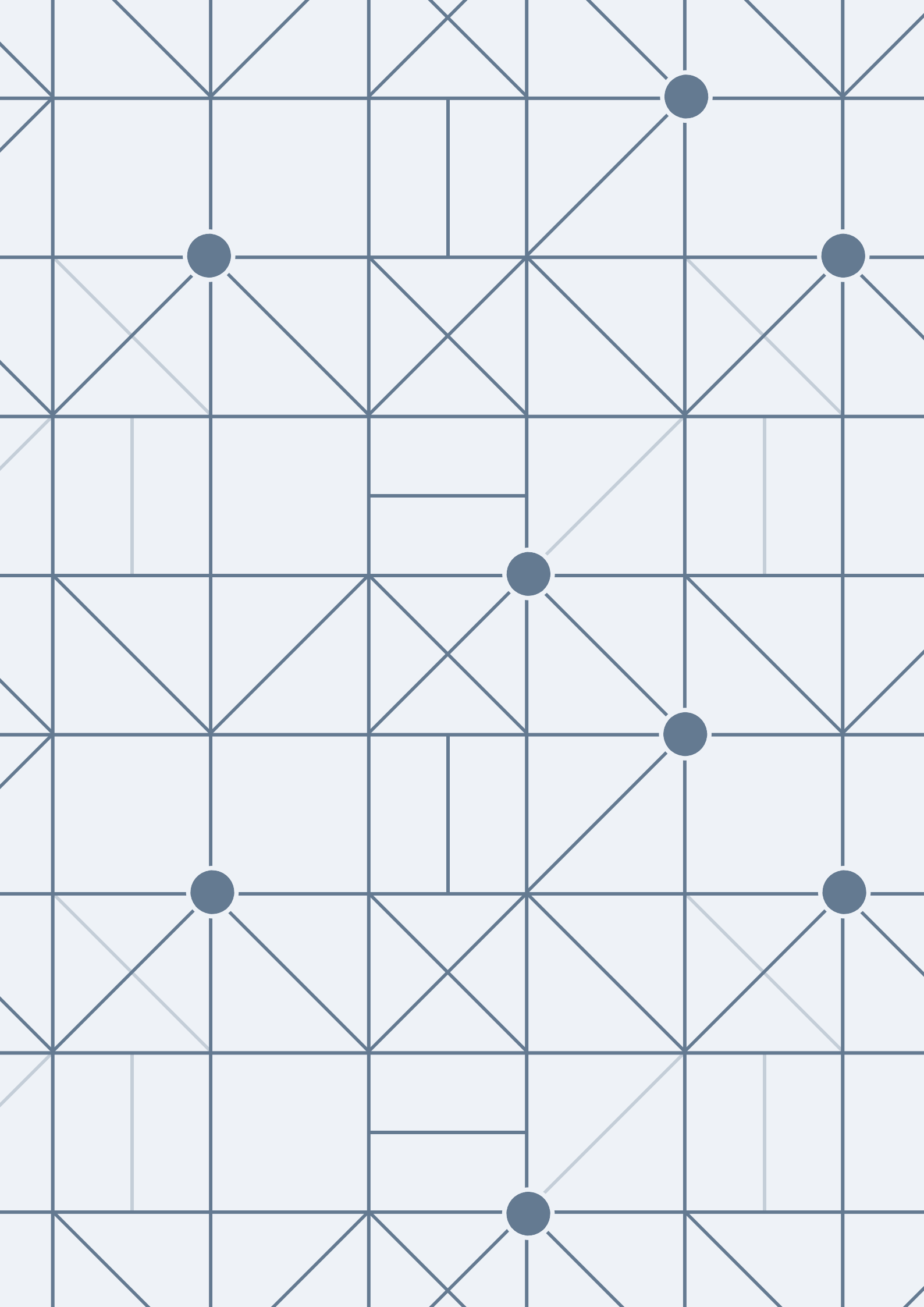




Micro canonical ensemble $\rightarrow$ simple - but this is not satisfied in any real system, it's always disordered.
 - then why do we study it? because it's the mathematical base to get the another ensembles.

POSTULATE OF AVERAGES
POSTULATE OF EQUAL A PRIORI PROBABILITY

The **POSTULATE OF AVERAGES** states that, for a system in thermal equilibrium, the time average of a physical quantity over a sufficiently long period is equal to the ensemble average for the same quantity in the corresponding ensemble.

The **POSTULATE OF EQUAL A PRIORI PROBABILITY** asserts that, in the absence of any additional information or constraints, all microstates consistent with a given macrostate are equally probable.

The two postulates represent the **building blocks** with which **we** will construct the assemblies of thermodynamic systems in equilibrium, starting with the simplest one: the **isolated system**.

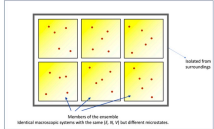
MICROCANONICAL ENSEMBLE: the members of the ensemble are macroscopic systems identical copies of the original, **ISOLATED** and **IN EQUILIBRIUM**.

ISOLATED SYSTEM: a system that does not interact thermally, mechanically, or materially with the surroundings; characterized by *adiabatic, rigid, and impenetrable walls*.

$$V = cte \quad N = cte \quad H = E$$

IN EQUILIBRIUM: there is **no time dependence** of any variable, and the macrostate can be described using Thermodynamics. The density matrix/density function will also be independent of time.

Ensemble of isolated and equilibrium systems: **macrostate** (N, V, E)



MICROCANONICAL ENSEMBLE: the members of the ensemble are macroscopic systems identical copies of the original, **ISOLATED** and **IN EQUILIBRIUM**.

In the **classical** case, a microstate of the system is characterised by a point in phase space, Γ .

This phase space encompasses all possible combinations of positions and momenta for each particle in the system, and each point within this space represents a unique microstate.

$$(q, p) = (q_1, p_1, \dots, q_s, p_s) \rightarrow 2s \text{ dimensions}$$

$\rho(\mathbf{q}, \mathbf{p}) \Rightarrow$ probability density for the system to be in the microstate $(\mathbf{q}, \mathbf{p})$ (consistent with the macrostate)

What probability (statistical weight) do we assign to each of these microstates?

- **ISOLATED:** a system that does not interact thermally, mechanically, or materially with the surroundings.

$$V = \text{const} \quad N = \text{const}$$

$$H(\mathbf{q}, \mathbf{p}) = E = \text{const} \quad (q, p) \equiv (q_1, \dots, q_{3N}, p_1, \dots, p_{3N})$$

- **IN EQUILIBRIUM:** the density function is stationary.

$$\frac{\partial \rho}{\partial t} = 0 \quad \longrightarrow \quad \rho(q, p) = \rho(H(q, p))$$

WHAT IS THE DENSITY FUNCTION FOR THE MICROCANONICAL ENSEMBLE?



With the help of the postulate of equal a priori probability, it is possible to construct this density function clearly and directly.

The density function for the microcanonical ensemble is **constant** for all microstates consistent with the macrostate, assigning them equal probability.

In other words, for a system in a microcanonical equilibrium, **the density function is uniform in phase space**, reflecting the equal a priori probability of occupying any allowed microstate.

An isolated system with energy E is an idealisation.



- In reality, we can never completely eliminate interactions with the external world.
- Any small error in determining the energy covers a very broad spectrum of microstates.
- According to the Heisenberg Uncertainty Principle, obtaining the exact energy would require a measurement of infinite duration.

Instead of specifying the energy, we will define an isolated system as one whose energies fall within the interval ΔE .

$$\begin{aligned} & [E, E + \Delta E] \\ & \Delta E \ll E \end{aligned}$$

Diagram illustrating a region Ω in the plane, bounded by a curve Γ . The region is shaded light blue. The boundary Γ is a closed curve. A point p is on the boundary. A small segment of the boundary is highlighted in red. The region is labeled $\Omega : \phi(c-\sigma), \phi(c)$. The boundary is labeled $\Gamma : \phi(c)$. The region is labeled $H(q,p) = E + \Delta \phi(E+\Delta)$. The boundary is labeled $H(q,p) = E$. The boundary is labeled $\phi(E) = \text{amount of area that lies in the boundary p.e.}$. The boundary is labeled $-4C1C$.

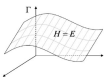
$$\Omega(E; \Delta E; N, V) = \Omega(E) = \phi(E + \Delta E) - \phi(E) \approx \frac{\partial \phi}{\partial E} \Delta E = \omega(E) \Delta E$$

We'll see that ΔF is not relevant.

Le type approximation

$$S(E) \approx \frac{\partial S}{\partial E} \Delta E = \frac{1}{T} \Delta E \rightarrow \frac{1}{T} = \frac{\partial S}{\partial E} \rightarrow \text{number of microstates in } \Delta E$$

1. Any average over this ensemble must take into account that all mechanical states $(\mathbf{q}, \mathbf{p})$ for which $H \neq E$ cannot contribute to the mean value since they do not conserve energy.



2. ρ is a function **exclusively** of H .
3. **Equal a priori** probability for all microstates with $H = E$.
4. **Normalisation** condition: $\int \rho(\mathbf{q}, \mathbf{p}) d\mathbf{q}d\mathbf{p} = 1$

$$\rho(q, p) = \frac{1}{\omega(N, V, E)} \delta(E - H)$$

$$\omega(N, V, E) = \int \delta(E - H(q, p)) dq dp$$

Disc delta excludes
non-compatible

$$w(\text{MVE}) = \frac{\partial d}{\partial \epsilon} = \int \delta(\epsilon - \eta) \, d\epsilon \, d\eta \quad \text{mit } \delta(\epsilon - \eta) = \begin{cases} 1 & \text{falls } \epsilon = \eta \\ 0 & \text{sonst} \end{cases}$$

$$g(\xi, \eta) = g(H) = \frac{1}{\omega} \delta(c-H)$$

$$p(x, \bar{x}) = \begin{cases} 1/2 & \text{exakte} \\ 0 & \text{sonst} \end{cases}$$

$$\rho(q,p) = \frac{1}{\omega(N,V,E)} \delta(E - H)$$

$$\omega(N,V,E) = \int \delta(E - H(q,p)) dq dp$$

These expressions for the classical density function are not entirely correct

We need a correction factor to adjust dimensions and avoid issues with entropy

A. This gives a volume, not a number of states!
It needs correction → dividing by the "unit volume", h^3

B. If particles are identical, swapping two particles does not give a new state.
Classical integral counts them as different → **overcounting**
Divide by $N!$ → correct number of states

→ without this correction, the entropy in the real world would be a massive overestimate. This means that for N not even that

→ h^3 → $4\pi p^2 h$, the volume that encloses a single state then volume $\frac{4\pi p^2 h}{2\pi} = h$ → particle

→ $N!$ → N particles

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Final expressions

$$\rho(q,p) = \frac{1}{h^3 N!} \frac{1}{\omega(N,V,E)} \delta(E - H)$$

$$\omega(N,V,E) = \frac{1}{h^3 N!} \int \delta(E - H(q,p)) dq dp$$

Mean value of the dynamic function $a(q,p)$

$$\langle a(q,p) \rangle = \frac{1}{h^3 N!} \frac{1}{\omega(N,V,E)} \int a(q,p) \delta(E - H(q,p)) dq dp$$

$$\langle a(q,p) \rangle = \int a(q,p) \rho(q,p) dq dp = \frac{1}{N!} \frac{1}{\omega(N,V,E)} \int a(q,p) \delta(E - H(q,p)) dq dp$$

Ω is the most important quantity in the microcanonical ensemble.

From Ω , and its dependence on the variables N , V , and E , the complete thermodynamics of the system can be deduced.



What is the connection between Ω and thermodynamics?